

HILAL MUHAMAD ABDUL GANI

Bandung, Jawa Barat | 081563955598 | hilalabdulgani@gmail.com
[linkedin.com/in/hilal-muhamad](https://www.linkedin.com/in/hilal-muhamad) | <https://hilalmuhamad.my.id/>

SUMMARY

Informatics Engineering graduate with hands-on full-stack development and network infrastructure experience across industry roles, government, and independent projects. Proficient in HTML, CSS, JavaScript, PHP, Laravel, FastAPI, React, and Flutter, with strong database skills in MySQL, PostgreSQL, and SQL Server. Experienced in mobile application architecture, enterprise-scale data system design, and network performance optimization. Delivered measurable impact across diverse technical domains by successfully architecting production-ready systems. A rapid learner, adaptable across new tools, and committed to delivering scalable software solutions in fast-paced environments.

WORK EXPERIENCE

BLSDM Komdigi Bandung

March 2026 – June 2026

Information Technology Support Intern

- Managed network infrastructure by designing a scalable, hierarchical topology using enterprise-grade Ruijie and Linksys access points. Enabled multi-floor expansion without disrupting daily operations.
- Conducted network troubleshooting and diagnostics through systematic wireless site surveys using NetSpot and WiFi Analyzer for signal coverage mapping and interference detection.
- Performed latency, throughput, and packet-loss testing using Speedtest and PingPlotter, resulting in a stable, low-latency, and high-throughput wireless connection across the facility.
- Collaborated with the IT team on network configuration planning, using Packet Tracer and GNS3 to simulate VLAN segmentation and switch configurations prior to production deployment, reducing configuration errors and minimizing rollout risk in a live environment.

Maqdis Academy

June 2026 – September 2026

Software Engineer Intern

- Developed core mobile UI modules in Flutter including a tajwid/recitation search feature, the Maqdis Academy training program directory, and a custom in-app audio player and implemented Background Playback via native background services to ensure uninterrupted Quran recitation audio when the app is minimized or the screen is locked.
- Contributed to the E-Course feature, integrating self-paced Quran reading lessons with a 1-on-1 private class registration flow for certified Islamic teachers (ustadz/ustadzah), and implemented REST API integrations with input validation to support hafalan recording uploads, private-class progress tracking, and student data synchronization.
- Implemented on-device local storage and caching for audio metadata and hafalan progress history, enabling stable offline playback and reducing dependency on constant network connectivity.
- Conducted functional and API integration testing using Postman and mobile debugging tools to validate hafalan data consistency, audio playback performance, and prayer/ibadah reminder push notification flows.

I Channel TV Bandung

April 2022 – June 2022

Assistant Producer Intern

- Managed daily television production workflows alongside the Senior Producer, focusing on structured scheduling and strict adherence to broadcasting quality standards.
- Coordinated cross-functional communication between the creative, technical, and talent teams, ensuring the seamless execution of project timelines.

PROJECTS

Tracer Study Data Warehouse & OLAP Analytics System

Fullstack Developer

https://github.com/hilalmuhamad/tugas_akhir.git

- Designed and implemented a Star Schema Data Warehouse comprising centralized fact tables and conformed dimensions, integrated via an automated ETL (Extract, Transform, Load) pipeline that maintained relational integrity with zero orphan records.
- Implemented a Hybrid Transactional/Analytical Processing (HTAP) environment using PostgreSQL and Laravel, successfully isolating real-time operational data (OLTP) from the analytical data warehouse to ensure zero performance degradation during complex aggregations.
- Built and developed robust OLAP functionalities (Roll-up, Slice, Dice, Drill-down) using advanced SQL queries and FILTER clauses, enabling dynamic, multi-dimensional analysis of alumni employability metrics (e.g., vertical/horizontal alignment, waiting periods) for institutional KPI dashboards.
- Optimized database scalability and query performance through strategic schema separation (public, staging, dwh), table partitioning by cohort year, and the implementation of Block Range Indexes (BRIN) on temporal dimensions.

Focus Talk – Android-Based Smart Intervention & Micro-Learning

Nov 2025 – Feb 2026

Mobile Developer

https://github.com/rdwnsyh/focustalk_app.git

- Built a full-stack Android intervention system using Flutter (client) and FastAPI/Python (backend), designed to reduce digital procrastination among students through real-time behavioral intervention.
- Engineered a Heuristic Rule-Based detection engine using Android's UsageStatsManager to monitor foreground app activity in real time and accurately classify distraction-triggering applications (e.g., Instagram, TikTok).
- Developed a Background Service leveraging System Alert Window (WindowManager) to overlay non-blocking English micro-learning quizzes, requiring quiz completion as a soft-gate before accessing distraction apps.
- Designed a FastAPI backend to synchronize user scores, profiles, and a leaderboard system, adding a gamification layer that incentivizes consistent engagement with the learning content.

Maggot Cycle – Distributed Backend for Organic Waste Reporting System

Jan 2026

Backend/Infra Engineer

- Architected a distributed microservices backend using Golang (Fiber) and Podman containerization, implementing horizontal scaling with dual backend instances behind an Nginx Load Balancer to handle high-concurrency reporting traffic from hundreds of B2B partner kitchens under a national food-distribution program.
- Designed a PostgreSQL Master-Slave replication architecture and disaster recovery strategy, achieving near-zero data loss (RPO) and sub-1-hour recovery time (RTO), validated through concurrent load testing that maintained a 0% error rate.
- Implemented multi-layer security hardening JWT-based API authentication, Cloudflare WAF, Suricata IDS/IPS, Fail2Ban, and Tailscale VPN for private server access — verified through Nmap-based penetration testing confirming zero exposed internal ports.

MediTech – Integrated Digital Health Management System

Jun 2025 – Aug 2025

Fullstack Developer

<https://github.com/hilalmuhamad/app-klinik-kita.git>

- Developed the back-end architecture using Node.js and Express.js (MongoDB), focusing on secure medical data management and real-time telemedicine capabilities.
- Built a multi-platform ecosystem featuring a React.js web dashboard for hospital administration and a Flutter mobile application for seamless patient access.
- Secured sensitive healthcare information by integrating JSON Web Tokens (JWT) for authentication and Role-Based Access Control (RBAC).

EDUCATION

Universitas Logistik dan Bisnis Internasional (Poltekpos)

Bandung, Indonesia | 2023 – 2026

Diploma (D3) in Informatics Engineering | GPA: 3.72/4.00

- Participated in a Community Service Program (PKM) at SMK Negeri 2 Cimahi focused on improving web programming skills through a Laravel bootcamp.
- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (OOP), Web & Mobile Development, Database Management Systems (SQL/NoSQL), Software Engineering.

- Final Project: Designed and developed an Alumni Tracer Study web application using OLAP-based analytics (star schema data warehouse) with a React.js/TailAdmin frontend and Laravel/PostgreSQL backend, enabling institutional monitoring of alumni employment alignment and career trajectories.

TRAININGS AND CERTIFICATIONS

BNSP <i>Teknisi Muda Jaringan Komputer</i>	Sep 2026
Coursera <i>Foundations of Cybersecurity</i>	Jul 2026
SAP <i>Introduction to the ABAP Workbench</i>	Jul 2025
Dicoding <i>Back-End Development with JavaScript</i>	Sep 2024
Dicoding <i>Cloud Practitioner Essentials</i>	Jun 2024

ORGANIZATIONAL EXPERIENCE

Industrial Visit Committee PT Computrade Teknologi Internasional <i>Project Officer</i>	Aug 2025
• Directed a committee of 24 members to organize a strategic industrial visit, serving as the primary liaison between the university and corporate representatives. • Managed the event timeline, logistics, and correspondence, ensuring a seamless experience for 76 student participants.	

HIMATIF (Informatics Engineering Student Association) <i>Staff Public Relations</i>	March 2025 - Present
• Delivered an internal training session on sponsorship prospecting strategies to 10–30 HIMATIF committee members, sharing practical frameworks for identifying and approaching potential corporate sponsors for student-led events. • Initiated and managed the "Safari Himpunan" (Comparative Study programs) with external university student associations. • Acted as the primary liaison between the student association and corporate sponsors, actively managing professional communications and negotiating strategic partnerships to support student-led events.	

K-Radio ULBI <i>Head of Media Design Division</i>	March 2025 - Present
• Led a creative team in producing digital content, resulting in a consistent visual identity across all social media platforms. • Oversaw the design workflow for campus event promotions, increasing audience engagement through visually compelling graphics.	

LANGUAGES & SKILLS

- **Mobile & Frontend:** Flutter, React.js, BLoC, Clean Architecture
- **Backend & Programming:** JavaScript, PHP, Python, Node.js/Express, Laravel, SQL
- **Databases:** MySQL, PostgreSQL, MongoDB, SQLite
- **Tools:** Git, Postman, VS Code, Figma
- **Networking & Infrastructure:** TCP/IP, VLAN Segmentation, Load Balancing, System Hardening, WAF (Cloudflare), Nginx, Tailscale
- **Containerization:** Podman
- **Soft Skills:** Strategic Communication, Cross-Functional Leadership
- **Languages:** Indonesian (Native), English (Professional Working Proficiency)